

```
mysql> CREATE TABLE Employees (  
->     emp_id INT PRIMARY KEY,  
->     name VARCHAR(100),  
->     department_id INT,  
->     salary DECIMAL(10,2)  
-> );  
Query OK, 0 rows affected (0.05 sec)
```

```
mysql>  
mysql> CREATE TABLE Departments (  
->     department_id INT PRIMARY KEY,  
->     department_name VARCHAR(100)  
-> );  
Query OK, 0 rows affected (0.04 sec)
```

```
mysql>  
mysql> CREATE TABLE Projects (  
->     project_id INT PRIMARY KEY,  
->     project_name VARCHAR(100),  
->     emp_id INT  
-> );  
Query OK, 0 rows affected (0.04 sec)
```

```
mysql>  
mysql> INSERT INTO Departments (department_id, department_name) VALUES  
-> (1, 'Human Resources'),  
-> (2, 'Engineering'),  
-> (3, 'Marketing'),  
-> (4, 'Finance'),  
-> (5, 'Operations');  
Query OK, 5 rows affected (0.02 sec)  
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql>  
mysql>  
mysql>  
mysql> INSERT INTO Employees (emp_id, name, department_id, salary) VALUES  
-> (101, 'Arjun', 1, 50000),  
-> (102, 'Bhavana', 2, 65000),  
-> (103, 'Charan', 2, 72000),  
-> (104, 'Divya', 3, 45000),  
-> (105, 'Eswar', NULL, 40000),  
-> (106, 'Farhan', 4, 55000),  
-> (107, 'Gowri', 2, 62000),  
-> (108, 'Harish', 5, 48000),  
-> (109, 'Ishita', NULL, 53000),  
-> (110, 'Jagan', 3, 60000);  
Query OK, 10 rows affected (0.01 sec)  
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql>
mysql>
mysql> INSERT INTO Projects (project_id, project_name, emp_id) VALUES
-> (201, 'Website Redesign', 102),
-> (202, 'Mobile Banking App', 103),
-> (203, 'Marketing Campaign', 104),
-> (204, 'Payroll System', 106),
-> (205, 'Backend Optimization', 107),
-> (206, 'Cloud Migration', 102),
-> (207, 'Customer Analytics', 110),
-> (208, 'Inventory System', NULL);
Query OK, 8 rows affected (0.01 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

```
mysql> select * from projects;
+-----+-----+-----+
| project_id | project_name      | emp_id |
+-----+-----+-----+
|          201 | Website Redesign  |      102 |
|          202 | Mobile Banking App |      103 |
|          203 | Marketing Campaign |      104 |
|          204 | Payroll System    |      106 |
|          205 | Backend Optimization |      107 |
|          206 | Cloud Migration   |      102 |
|          207 | Customer Analytics |      110 |
|          208 | Inventory System  |     NULL |
+-----+-----+-----+
8 rows in set (0.00 sec)
```

```
mysql> select * from departments;
+-----+-----+
| department_id | department_name |
+-----+-----+
|             1 | Human Resources |
|             2 | Engineering     |
|             3 | Marketing       |
|             4 | Finance         |
|             5 | Operations      |
+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> select * from employees;
+-----+-----+-----+-----+
| emp_id | name   | department_id | salary |
+-----+-----+-----+-----+
|      101 | Arjun  |             1 | 50000.00 |
|      102 | Bhavana |             2 | 65000.00 |
|      103 | Charan  |             2 | 72000.00 |
|      104 | Divya  |             3 | 45000.00 |
+-----+-----+-----+-----+
```

105	Eswar	NULL	40000.00
106	Farhan	4	55000.00
107	Gowri	2	62000.00
108	Harish	5	48000.00
109	Ishita	NULL	53000.00
110	Jagan	3	60000.00

10 rows in set (0.00 sec)

mysql> -- get all employees with there departments?

```
mysql> select e.emp_id,e.name,d.department_name
-> from employees e
-> inner join departments d
-> on e.department_id=d.department_id;
```

emp_id	name	department_name
101	Arjun	Human Resources
102	Bhavana	Engineering
103	Charan	Engineering
104	Divya	Marketing
106	Farhan	Finance
107	Gowri	Engineering
108	Harish	Operations
110	Jagan	Marketing

8 rows in set (0.00 sec)

mysql> -- get all employees who assigned to a project?

```
mysql> select e.emp_id,e.name, p.project_name
-> from employees e
-> inner join projects p
-> on e.emp_id=p.emp_id;
```

emp_id	name	project_name
102	Bhavana	Website Redesign
103	Charan	Mobile Banking App
104	Divya	Marketing Campaign
106	Farhan	Payroll System
107	Gowri	Backend Optimization
102	Bhavana	Cloud Migration
110	Jagan	Customer Analytics

7 rows in set (0.00 sec)

```
mysql> select e.emp_id,e.name
-> from employees e
-> inner join projects p
-> on e.emp_id=p.emp_id;
```

emp_id	name
102	Bhavana
103	Charan
104	Divya
106	Farhan
107	Gowri
102	Bhavana
110	Jagan

7 rows in set (0.00 sec)

```
mysql> select distinct e.emp_id,e.name from employees e inner join projects p on
e.emp_id=p.emp_id;
```

emp_id	name
102	Bhavana
103	Charan
104	Divya
106	Farhan
107	Gowri
110	Jagan

6 rows in set (0.00 sec)

```
mysql> select e.emp_id,e.name,d.department_name
-> from employees e
-> left join departments d
-> on e.department_id=d.department_id;
```

emp_id	name	department_name
101	Arjun	Human Resources
102	Bhavana	Engineering
103	Charan	Engineering
104	Divya	Marketing
105	Eswar	NULL
106	Farhan	Finance
107	Gowri	Engineering
108	Harish	Operations
109	Ishita	NULL
110	Jagan	Marketing

10 rows in set (0.00 sec)

```
mysql> select e.emp_id,e.name,p.project_name
-> from employees e
-> left join projects p
```

-> on e.emp_id=p.emp_id where project_name is NULL;

emp_id	name	project_name
101	Arjun	NULL
105	Eswar	NULL
108	Harish	NULL
109	Ishita	NULL

4 rows in set (0.00 sec)

mysql> -- All departments and the employees in them

mysql> select d.department_name,e.name
-> from departments d
-> right join employees e
-> on d.department_id = e.department_id;

department_name	name
Human Resources	Arjun
Engineering	Bhavana
Engineering	Charan
Marketing	Divya
NULL	Eswar
Finance	Farhan
Engineering	Gowri
Operations	Harish
NULL	Ishita
Marketing	Jagan

10 rows in set (0.00 sec)

mysql> select e1.name as employee1,e2.name as employee2,e1.department_id from
employees e1 join employees e2 on e1.department_id=e2.department_id and
e1.emp_id!=e2.emp_id;

employee1	employee2	department_id
Gowri	Bhavana	2
Charan	Bhavana	2
Gowri	Charan	2
Bhavana	Charan	2
Jagan	Divya	3
Charan	Gowri	2
Bhavana	Gowri	2
Divya	Jagan	3

8 rows in set (0.00 sec)

mysql> select e.name,d.department_name from employees e left join departments d

```

-> on e.department_id=d.department_id
-> union
-> select e.name,d.department_name from employees e right join departments d
-> on e.department_id=d.department_id;

```

name	department_name
Arjun	Human Resources
Bhavana	Engineering
Charan	Engineering
Divya	Marketing
Eswar	NULL
Farhan	Finance
Gowri	Engineering
Harish	Operations
Ishita	NULL
Jagan	Marketing

10 rows in set (0.00 sec)

```

mysql> select d.department_name,count(e.emp_id) as total_employees
-> from departments d
-> left join employees e on d.department_id=e.department_id
-> group by d.department_name;

```

department_name	total_employees
Human Resources	1
Engineering	3
Marketing	2
Finance	1
Operations	1

5 rows in set (0.00 sec)

```

mysql> select d.department_name,sum(e.salary) as total_salary
-> from departments d
-> left join employees e on d.department_id=e.department_id
-> group by d.department_name;

```

department_name	total_salary
Human Resources	50000.00
Engineering	199000.00
Marketing	105000.00
Finance	55000.00
Operations	48000.00

5 rows in set (0.00 sec)

```
mysql> select e.name,count(p.project_id) as projects_count
-> from employees e
-> left join projects p on e.emp_id = p.emp_id
-> group by e.name;
```

name	projects_count
Arjun	0
Bhavana	2
Charan	1
Divya	1
Eswar	0
Farhan	1
Gowri	1
Harish	0
Ishita	0
Jagan	1

10 rows in set (0.00 sec)

```
mysql> select e.name,d.department_name,e.salary from employees e
-> join departments d on e.department_id=d.department_id
-> order by e.salary;
```

name	department_name	salary
Divya	Marketing	45000.00
Harish	Operations	48000.00
Arjun	Human Resources	50000.00
Farhan	Finance	55000.00
Jagan	Marketing	60000.00
Gowri	Engineering	62000.00
Bhavana	Engineering	65000.00
Charan	Engineering	72000.00

8 rows in set (0.00 sec)

```
mysql> select d.department_name,count(e.emp_id) as employee_count
-> from departments d
-> left join employees e on d.department_id=e.department_id
-> group by d.department_name
-> order by employee_count desc;
```

department_name	employee_count
Engineering	3
Marketing	2
Human Resources	1
Finance	1
Operations	1

```
+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> select d.department_name,count(e.emp_id) as employee_count
-> from departments d
-> join employees e on d.department_id=e.department_id
-> group by d.department_name
-> having count(e.emp_id) > 1
-> order by employee_count desc;
```

```
+-----+-----+
| department_name | employee_count |
+-----+-----+
| Engineering    | 3              |
| Marketing      | 2              |
+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> select d.department_name,sum(e.salary) as total_salary
-> from departments d
-> left join employees e on d.department_id=e.department_id
-> group by d.department_name
-> having sum(e.salary) > 50000
-> order by total_salary desc;
```

```
+-----+-----+
| department_name | total_salary |
+-----+-----+
| Engineering    | 199000.00   |
| Marketing      | 105000.00   |
| Finance        | 55000.00    |
+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> notee
```